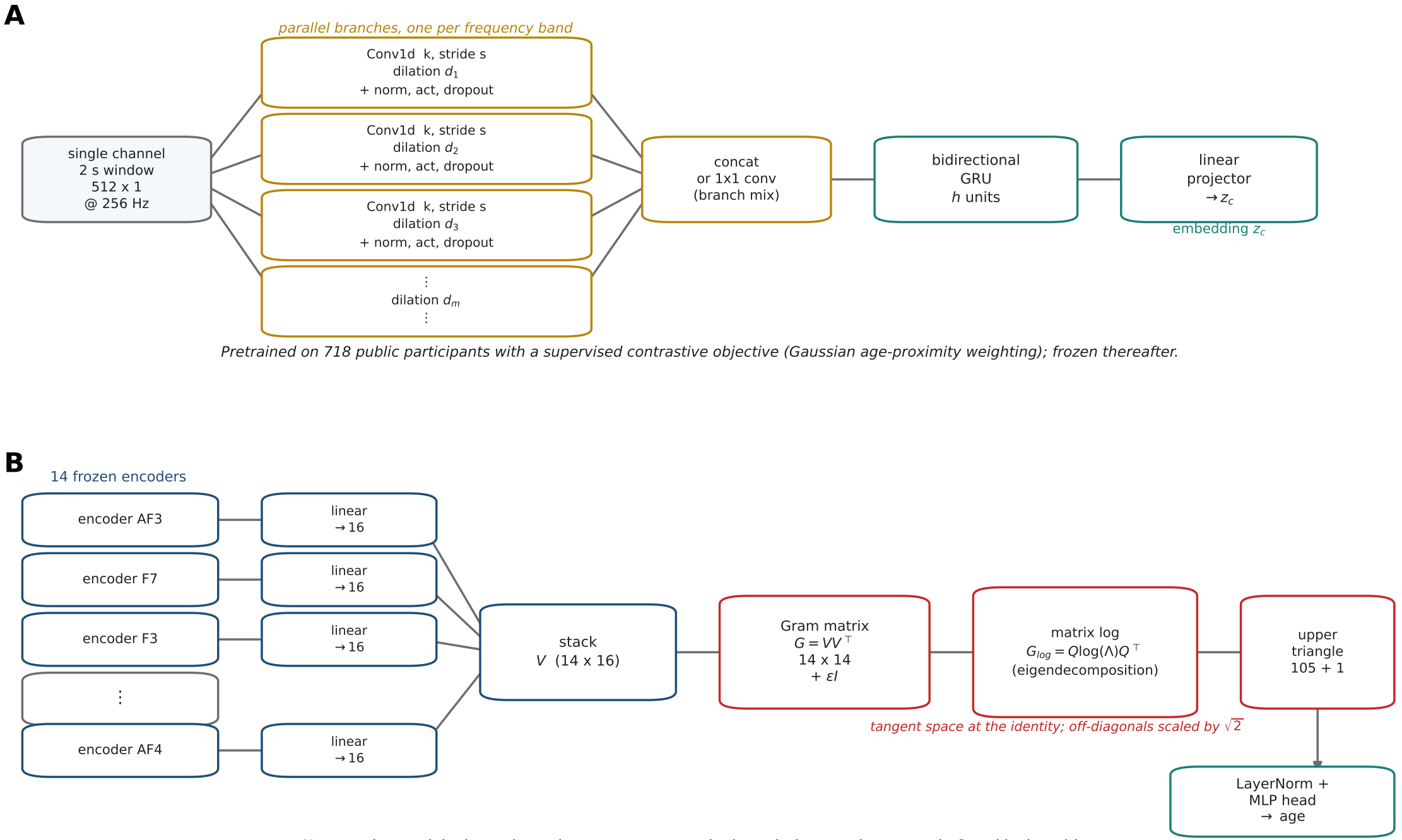
A
parallel branches, one per frequency band
single channel
2 s window
512 x 1
@ 256 Hz
Conv1d k, stride s
dilation d_1
+ norm, act, dropout
Conv1d k, stride s
dilation d_2
+ norm, act, dropout
Conv1d k, stride s
dilation d_3
+ norm, act, dropout
⋮
dilation d_m
⋮
concat
or 1x1 conv
(branch mix)
bidirectional
GRU
h units
linear
projector
$\rightarrow z_c$
embedding z_c
Pretrained on 718 public participants with a supervised contrastive objective (Gaussian age-proximity weighting); frozen thereafter.
B
14 frozen encoders
encoder AF3
encoder F7
encoder F3
⋮
encoder AF4
linear
$\rightarrow 16$
linear
$\rightarrow 16$
linear
$\rightarrow 16$
linear
$\rightarrow 16$
stack
V (14 x 16)
Gram matrix
$G = VV^\top$
14 x 14
$+ \varepsilon I$
matrix log
$G_{log} = Q\log(\Lambda)Q^\top$
(eigendecomposition)
tangent space at the identity; off-diagonals scaled by $\sqrt{2}$
upper
triangle
105 + 1
LayerNorm +
MLP head
→ age
No attention module: inter-channel structure enters only through the covariance matrix G and its logarithm.